

Regression

~~Linear (Ridge)~~ Regression. General regression Task.

let $y_i \in \mathbb{R}$, $i=1,2,\dots,N$ be the response variables.

let $x_i \in \mathbb{R}^d$, $i=1,2,\dots,N$ be the predictors.

$\{(x_i, y_i)\}$ form the training set. ~~we wish to~~ The regression wish to find a ~~stat~~ $f: \mathbb{R}^d \rightarrow \mathbb{R}$ s.t. $f(x_i) \approx y_i$ for test set.

- function form (parametric methods)
assum
- assign ~~metric~~ ^{metric} ~~measure~~ for $f(x_i) \approx y_i$ (loss function).

Linear (Ridge) Regression.

f takes linear form $\Rightarrow y = w^T x + b$, $w \in \mathbb{R}^d$, $b \in \mathbb{R}$.

or $y = w^T x$ if we extend $x \rightarrow \begin{pmatrix} x_1 \\ \vdots \\ x_d \\ 1 \end{pmatrix}$

metric: least square error.

$$\min_{w \in \mathbb{R}^d} \sum_i (y_i - \langle w, x_i \rangle)^2 \rightarrow \text{error between predicted and actual.}$$

$$\text{or } \|Y - Xw\|_2^2 \text{ if we denote } Y = \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix} \in \mathbb{R}^N, X = \begin{pmatrix} x_1^T \\ \vdots \\ x_N^T \end{pmatrix} \in \mathbb{R}^{N \times d}.$$

regularization (ridge regression)

in some case, the f could be overfitting (~~use~~ ^{noise} matching)

this can be reduced by adding regularization term in

the loss function, as a penalty for too large coefficient.

$$\Rightarrow \min_{w \in \mathbb{R}^d} \|Y - Xw\|_2^2 + \lambda \|w\|_2^2, \lambda > 0$$

from optimization theory, we can show that the minimization problem constructed is a strictly convex function, and has local minimum as global minimum.

Solution: $w^* = \arg\min ||Xw - y||_2^2 + \lambda ||w||_2^2 \quad \lambda > 0$

$$= (X^T X + \lambda I)^{-1} X^T y$$

proof sketch: find the gradient $\frac{\partial \mathcal{L}}{\partial w_i} = 0$, and Hessian $\frac{\partial^2 \mathcal{L}}{\partial w_i \partial w_j}$.
 this shouldn't be hard as $\mathcal{L}$ is in quadratic form.

Remarks — choice of regularizer.

- $\lambda ||w||_1 = \lambda \sum_i |w_i|$ — for sparsened error vector.
- $||w||_F = \sqrt{\sum_i \sum_j w_{ij}^2}$ — for low rank matrix.

Remarks — multiple labels predictors.

if $y_i \in \mathbb{R}^p$ as multiple labels, then we simply extend.

$$y_i = W x_i, \quad W \in \mathbb{R}^{p \times d}.$$

$$\Rightarrow \begin{pmatrix} y_1^T \\ \vdots \\ y_p^T \end{pmatrix} = \begin{pmatrix} w_{11}, w_{12}, \dots, w_{1d} \\ \vdots \\ w_{p1}, w_{p2}, \dots, w_{pd} \end{pmatrix} \begin{pmatrix} x_1^T \\ \vdots \\ x_d^T \end{pmatrix}$$

$$\text{and } Y = \begin{pmatrix} y_1^T \\ \vdots \\ y_N^T \end{pmatrix}, \quad X = \begin{pmatrix} x_1^T \\ \vdots \\ x_N^T \end{pmatrix}, \quad \mathcal{L} = ||XW^T - Y||_2^2 + \lambda ||w||_2^2.$$

Polynomial and Ridge Regression

It needs not to be linear, as long as it is linear.
Parametrized.

Example: Consider 3rd-order ~~polynomial~~ polynomial.
fit

$$y = w_3^T x^3 + w_2^T x^2 + w_1^T x = (w_3^T : w_2^T : w_1^T) \begin{pmatrix} x^3 \\ x^2 \\ x \end{pmatrix}$$

which share the similar form as ^{single} linear regression.

Kernel regression.

The kernel trick can be applied to the features x . Suppose $\phi: \mathbb{R}^p \rightarrow \mathbb{R}^p$ as a feature map which transfers the original data to $(\phi(x_1), y_1), (\phi(x_2), y_2), \dots, (\phi(x_n), y_n) \in \mathbb{R}^p \times \mathbb{R}$.

Then perform linear ridge regression in the transformed feature space.

$$\min_{w \in \mathbb{R}^p} \sum_i (w^T \phi(x_i) - y_i)^2 + \lambda \|w\|_2^2.$$

From before, we know the optimal solution.

$$w^* = (\Phi^T \Phi + \lambda I_p)^+ \Phi^T y, \quad \Phi = \begin{pmatrix} \phi(x_1)^T \\ \phi(x_2)^T \\ \vdots \\ \phi(x_n)^T \end{pmatrix} \in \mathbb{R}^{n \times p}.$$

Just as before, directly calculate kernel ϕ and w^* is very costly when p is large. Want to apply kernel trick as before.

Lemma (Woodbury matrix identity).

$$(\Phi^T \Phi + \lambda Z_p)^T \Phi^T = \Phi^T (\Phi \Phi^T + \lambda Z_p)^T.$$

proof: the SVD of $\Phi = U \Sigma U^T$, then

$$\begin{aligned} LHS &= ((U \Sigma^T U^T)(U \Sigma V^T) + \lambda Z_p)^T V \Sigma^T U^T \\ &= (V \Sigma^T \Sigma V^T + \lambda Z_p V V^T)^T V \Sigma^T U^T \\ &= (V (\Sigma^T \Sigma + \lambda Z_p) V^T)^T V \Sigma^T U^T \\ &= V (\Sigma^T \Sigma + \lambda Z_p)^T V^T V \Sigma^T U^T = V (\Sigma^T \Sigma + \lambda Z_p)^T \Sigma U^T. \end{aligned}$$

$$\begin{aligned} RHS &= V \Sigma^T U^T ((U \Sigma V^T)(U \Sigma^T U^T) + \lambda Z_p)^T \\ &= V \Sigma^T U^T (U \Sigma \Sigma^T U^T + \lambda Z_p U U^T)^T \\ &= V \Sigma^T U^T (U (\Sigma \Sigma^T + \lambda Z_p) U^T)^T \\ &= V \Sigma^T U^T U (\Sigma \Sigma^T + \lambda Z_p)^T U^T \\ &= V \Sigma^T (\Sigma \Sigma^T + \lambda Z_p)^T U^T \end{aligned}$$

Detailed calculation on $\Sigma^T (\Sigma \Sigma^T + \lambda Z_p)^T \Sigma$.

show that they both equal to $p \times n$ pseudo diagonal.

matrix $\frac{\Phi_i^2}{\Phi_i^2 + \lambda}$.

thus we have $LHS = RHS$.

for To make prediction $y = w^*^T \phi(x) = \phi^T \Phi^T (\Phi \Phi^T + \lambda Z_p)^T$
 $= K_x^T (K + \lambda Z_n)^T y.$

where $K_{ij} = \phi(x_i)^T \phi(x_j)$ is the kernel matrix, and.

$$= K(x_i, x_j) \quad K_{x_i, x}^T = \phi^T(x_i) \phi(x) = K(x_i, x).$$

proof: $\Phi \Phi^T = \begin{pmatrix} \phi(x_1)^T \\ \phi(x_2)^T \\ \vdots \\ \phi(x_n)^T \end{pmatrix} \begin{pmatrix} \phi(x_1) & \dots & \phi(x_n) \end{pmatrix}$

$$= \begin{pmatrix} \phi(x_1)^T \phi(x_1) & \phi(x_1)^T \phi(x_2) & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

$$= \begin{pmatrix} K(x_1, x_1) & K(x_1, x_2) & \dots \\ \vdots & \ddots & \ddots \end{pmatrix}$$

and

$$\Phi \phi(x) = \begin{pmatrix} \phi(x_1)^T \\ \vdots \\ \phi(x_n)^T \end{pmatrix} \phi(x) = \begin{pmatrix} \phi(x_1)^T \phi(x) \\ \vdots \\ \phi(x_n)^T \phi(x) \end{pmatrix} = \begin{pmatrix} K(x_1, x) \\ \vdots \\ K(x_n, x) \end{pmatrix}$$

Remark: using kernel trick, the kernel ridge regression cost.

• ^{pre}compute $K \in \mathbb{R}^{n \times n}$, cost $O(n^2)$ kernel evals.

• $(K + \lambda I_n)^T y$ precompute, cost $O(n^2)$ flops & $O(n)$ storage.

• compute Kx , cost $O(n)$ kernel evals

• compute $K^T (K + \lambda I_n)^T y$, cost $O(n)$ flops